Inverse hyperbolic trig functions

Key Points

As hyperbolic trig functions are exponentials the inverses are logarithmic

$$\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2+1}\right)$$

$$\operatorname{arccosh} x = \ln\left(x \pm \sqrt{x^2-1}\right)$$

$$\operatorname{arctanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

Basic Skills

Q1. Use the exponential definitions solve $\cosh x = 3$

Q2. State the exact value of

a) $\operatorname{arcsinh} 2$

b) $\operatorname{arctanh}\left(-\frac{1}{2}\right)$

Q3. Explain why there are no solutions to $\tanh x = 3$

Q4. Sketch the graphs stating the domain and range

a) $y = \operatorname{arcsinh} x$

b) $y = \operatorname{arccosh} x$

c) $y = \operatorname{arctanh} x$

Competency Skills

Q5. Solve the following by using the inverse formula

a) $\sinh x = -4$ **b)** $\cosh x = 3$ **c)** $\tanh x = -0.1$

Q6. Prove the following formulae

a) $\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2+1}\right)$

b) $\operatorname{arccosh} x = \ln\left(x \pm \sqrt{x^2-1}\right)$

c) $\operatorname{arctanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$

Q7. Solve the equation $\sinh(2x-1) = 5$

Q8. Solve the equation $\cosh^2 x = 4$

Advanced Skills

Q9. Show that $\operatorname{arccosh} x < \operatorname{arcsinh} x$ for $x > 1$

Q10. Solve the equation $x = \sinh(\operatorname{arccosh} 4x)$

Q11. Solve the equation $\sinh^2 x = 3(\cosh x - 1)$

Q12. For the equation $a\cosh x - b\sinh x = c$

a) Solve it when $a = 2, b = 3, c = 0$

b) Find the constants such that the equation has exactly one real solution

Extension

Q13. Show that the equation $\cosh x = x$ has no solutions

Hint: Consider showing $\cosh x > x$